MOBILE COMPUTING LAB

PRACTICAL RECORD

PAPER CODE: **ETIT-452**

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Group: **8C7**

Branch: **CSE**

S.No	EXPERIMENT NAME	DATE	MARKS					Total Marks	Signature
			R1	R2	R3	R4	R5		
1.	Write a WML program to print a formatted Text on the mobile Screen using various tags.								
2.	Write a WML program to connect multiple cards from same desk.								
3.	Write WML program to display table with three columns Image name, Image and third column contain hyperlink to open another card.								
4.	Write a WML program to create a form with multiple options.								
5.	Write a WML program to use the time control and to trigger On pick event.								
6.	Write a WML script to find maximum out of two numbers with help of inbuilt function Lang.Max() and to find absolute value with help of inbuilt function Lang.abs()								
7.	Write a Program in NS3 to Simulate OLSR.								
8.	Write a Program in NS3 to Simulate AODV.								
9.	Write a Android Program to create list view, grid view and database connectivity								
10.	Make an application of using XML or Android from the following: Game,								

	Clock, calendar, Convertor, phone book, Text Editor.								
11.	Developing Android App such as E-Commerce based or E learning Based apps								
12.	Program to develop a calling application.(Android)								
13.	Program to develop a mailing application. (Android)								
14.	Program to design a multiple activity application. (Android)								

Experiment-9

Aim: Write an Android Program to create list view, grid view and database connectivity.

Code:

```
implementation 'androidx.recyclerview:recyclerview:1.2.1'
implementation 'androidx.gridlayout:gridlayout:1.0.0'
implementation 'androidx.cardview:cardview:1.0.0'
implementation 'androidx.sqlite:sqlite:2.2.0'
```

activity_lab_list.xml: Layout file for the main activity.
list_item_lab.xml: Layout file for the list/grid item.
DatabaseHelper.java: Helper class to manage the SQLite database.
LabListActivity.java: Java file for the main activity.

activity_lab_list.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:visibility="gone" />

    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="2"
        android:horizontalSpacing="10dp"
        android:verticalSpacing="10dp"
        android:stretchMode="columnWidth"
        android:visibility="visible" />

</RelativeLayout>
```

list_item_lab.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
```

```
    android:padding="16dp">

    <ImageView

    android:id="@+id/imageView"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:src="@drawable/ic_launcher_background" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_marginStart="16dp"
        android:textSize="18sp"
        android:text="Lab Name" />

</LinearLayout>
```

DatabaseHelper.java:

```
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

import java.util.ArrayList;
import java.util.List;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "labs.db";
    private static final int DATABASE_VERSION = 1;

    private static final String TABLE_NAME = "labs";
    private static final String COLUMN_ID = "id";
    private static final String COLUMN_NAME = "name";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String createTable = "CREATE TABLE " + TABLE_NAME + " (" +
                COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, " +
                COLUMN_NAME + " TEXT)";
        db.execSQL(createTable);
```

```
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public void addLab(String labName) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(COLUMN_NAME, labName);
        db.insert(TABLE_NAME, null, values);
        db.close();
    }

    public List<String> getAllLabs() {
        List<String> labs = new ArrayList<>();
        String selectQuery = "SELECT * FROM " + TABLE_NAME;
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor cursor = db.rawQuery(selectQuery, null);

        if (cursor.moveToFirst()) {
            do {
                String labName = cursor.getString(cursor.getColumnIndex(COLUMN_NAME));
                labs.add(labName);
            } while (cursor.moveToNext());
        }

        cursor.close();
        db.close();
        return labs;
    }
}
```

LabListActivity.java:

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;
import java.util.List;

public class LabListActivity extends AppCompatActivity {

    private ListView listView;
    private GridView gridView;
    private DatabaseHelper databaseHelper;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_lab_list);

        listView = findViewById(R.id.listView);
        gridView = findViewById(R.id.gridView);

        databaseHelper = new DatabaseHelper(this);
        populateLabs();
    }

    private void populateLabs() {
        List<String> labs = databaseHelper.getAllLabs();

        ArrayAdapter<String> adapter = new ArrayAdapter<>(this, R.layout.list_item_lab, R.id.textView,
labs);

        listView.setAdapter(adapter);
        gridView.setAdapter(adapter);
    }
}
```

OUTPUT

10:20
LisrView
Item1
SEND
Item2
SEND
Item3
SEND
Item4
SEND
Item5
SEND
Item6
SEND
8C7
Mobile Computing

Experiment-10

Aim: Make an application using XML or Android from the following: Game, Clock, calendar, Convertor, phone book, Text Editor.

Code:

MainActivity.java

```
package com.example.phoneBookapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    private Button phoneBookButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        phoneBookButton = findViewById(R.id.phoneBookButton);
        phoneBookButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(MainActivity.this, phoneBookActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

phoneBookActivity.java:

```
package com.example.phoneBookapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.EditText;

public class phoneBookActivity extends AppCompatActivity {

    private EditText editText;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_text_editor);

        editText = findViewById(R.id.editText);
    }
}
```

activity_main.xml (layout for the main activity):

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/phoneBookButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Text Editor" />

</LinearLayout>
```

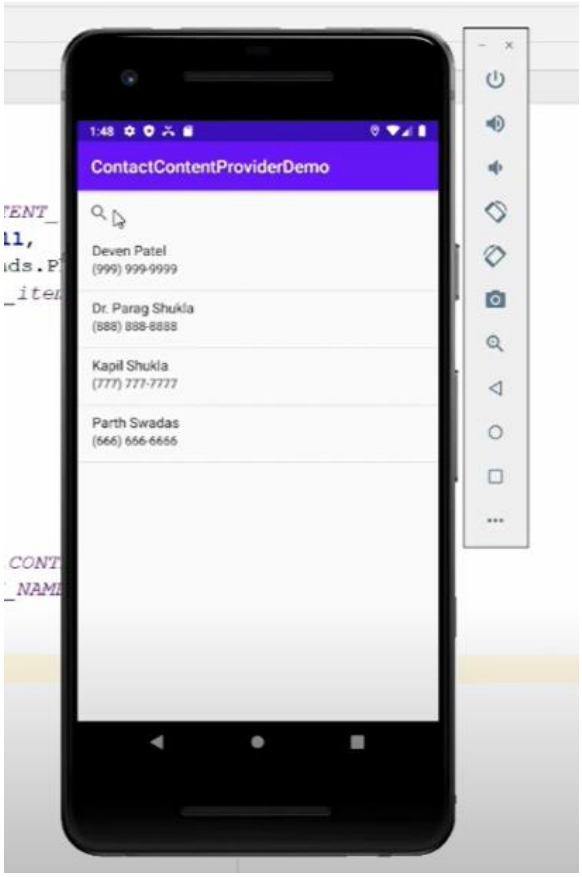
activity_text_editor.xml (layout for the Text Editor activity):

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".phoneBookActivity">

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="top|start"
        android:inputType="textMultiLine"
        android:minLines="10"
        android:hint="Start typing your text here..." />

</LinearLayout>
```

strings.xml (values resource file):

```
<resources>
    <string name="app_name">Text Editor App
```

OUTPUT



Experiment-11

Aim: Develop an android app such as E-Commerce based or E-learning.

Code:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/bgimage"
    tools:context=".HomeActivity">

    <androidx.appcompat.widget.Toolbar
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="#000"
        android:id="@+id/toolbar"
        android:elevation="4dp"
        app:layout_constraintBottom_toTopOf="@+id/table"
        android:theme="@style/ThemeOverlay.AppCompat.Dark.ActionBar"
        app:popupTheme="@style/ThemeOverlay.AppCompat.Light"/>

    <TableLayout
        android:id="@+id/table"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingHorizontal="10dp"
        app:layout_constraintTop_toBottomOf="@+id/toolbar">

        <TableRow>

 …..
        </TableRow>

        <TableRow>

            <TextView
                android:id="@+id/budgetTv"
                android:layout_height="wrap_content"
                android:layout_width="wrap_content"
                android:text="₹0"
                android:layout_weight="1"
                android:textColor="#fff"
                android:gravity="center"
```

```
            android:textStyle="bold"
            android:background="@drawable/borders"
            android:padding="5dp"
            android:textSize="12sp"/>

        <TextView
            android:id="@+id/todayTv"
            android:layout_height="wrap_content"
            android:layout_width="wrap_content"
            android:text="₹0"
            android:layout_weight="1"
            android:textColor="#fff"
            android:gravity="center"
            android:textStyle="bold"
            android:background="@drawable/borders"
            android:padding="5dp"
            android:textSize="12sp"/>

        <TextView
            android:id="@+id/weekTv"
            android:layout_height="wrap_content"
            android:layout_width="wrap_content"
            android:text="₹0"
            android:layout_weight="1"
            android:textColor="#fff"
            android:gravity="center"
            android:textStyle="bold"
            android:background="@drawable/borders"
            android:padding="5dp"
            android:textSize="12sp"/>
        <TextView
            android:id="@+id/savingsTv"
            android:layout_height="wrap_content"
            android:layout_width="wrap_content"
            android:text="₹0"
            android:layout_weight="1"
            android:textColor="#fff"
            android:gravity="center"
            android:textStyle="bold"
            android:background="@drawable/borders"
            android:padding="5dp"
            android:textSize="12sp"/>

    </TableRow>
</TableLayout>


            android:layout_gravity="fill"
            android:layout_rowWeight="1"
            android:layout_columnWeight="1"
```

```
    android:layout_margin="8dp"
    android:background="#fff">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center"
        android:layout_gravity="center">

    </LinearLayout>

</androidx.cardview.widget.CardView>
<androidx.cardview.widget.CardView
    android:id="@+id/weekCardView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_row="1"
    android:layout_column="0"
    android:layout_gravity="fill"
    android:layout_rowWeight="1"
    android:layout_columnWeight="1"
    android:layout_margin="8dp"
    android:background="#fff">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center"
        android:layout_gravity="center">

        <ImageView
            android:layout_width="60dp"
            android:layout_height="60dp"
            android:src="@drawable/week"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Week"
            android:textStyle="bold"
            android:textColor="#000"
            android:textAlignment="center"/>

    </LinearLayout>

</androidx.cardview.widget.CardView>
<androidx.cardview.widget.CardView
    android:id="@+id/monthCardView"
```

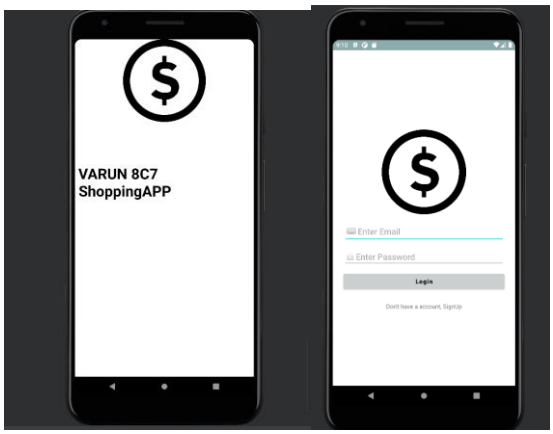
```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_row="1"
            android:layout_column="1"
            android:layout_gravity="fill"
            android:layout_rowWeight="1"
            android:layout_columnWeight="1"
            android:layout_margin="8dp"
            android:background="#fff">

            <LinearLayout
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:orientation="vertical"
                android:gravity="center"
                android:layout_gravity="center">


        </GridLayout>

        <com.google.android.material.floatingactionbutton.FloatingActionButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="bottom|end"
            app:fabSize="normal"
            android:id="@+id/fab"
            android:src="@drawable/ic_add"
            android:layout_margin="16dp"/>

    </androidx.coordinatorlayout.widget.CoordinatorLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

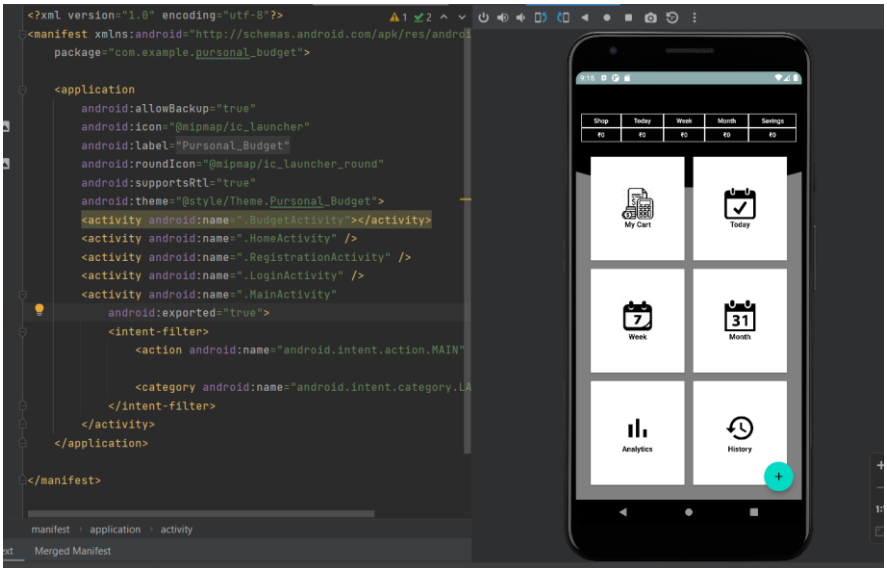
LabListActivity.java:

```
public class LoginActivity extends AppCompatActivity {
    EditText email,password;
    Button loginbtn;
    TextView signup;

    private FirebaseAuth mAuth;
    private ProgressDialog progressDialog;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);
        email=findViewById(R.id.email);
        password=findViewById(R.id.password);
```

```
signup=findViewById(R.id.signup);
loginbtn=findViewById(R.id.loginBtn);

mAuth=FirebaseAuth.getInstance();
progressDialog=new ProgressDialog(this);

signup.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent intent=new Intent(LoginActivity.this,RegistrationActivity.class);
        startActivity(intent);
    }
});

loginbtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String emailString=email.getText().toString();
        String passwordString=password.getText().toString();
        if(TextUtils.isEmpty(emailString)){email.setError("Email is required");}
        else if(TextUtils.isEmpty(passwordString)){password.setError("Email is required");}
         else{
            progressDialog.setMessage("login in progress");     //search more
            progressDialog.setCanceledOnTouchOutside(false);
            progressDialog.show();


mAuth.signInWithEmailAndPassword(emailString,passwordString).addOnCompleteListener(new
OnCompleteListener<AuthResult>() {
                @Override
                public void onComplete(@NonNull Task<AuthResult> task) {
                    if(task.isSuccessful()){
                        Intent intent=new Intent(LoginActivity.this,HomeActivity.class);
                        startActivity(intent);
                        finish();
                        progressDialog.dismiss();
                    }
                    else{
                        Toast.makeText(LoginActivity.this, task.getException().toString(),
Toast.LENGTH_SHORT).show();
                        progressDialog.dismiss();
                    }
```

OUTPUT


VARUN 8C7
ShoppingAPP
Enter Email
Enter Password
Login
Don't have a account, SignUp


<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/andro
package="com.example.pursonal_budget">
<application
android:allowBackup="true"
android:icon="@mipmap/ic_launcher"
android:label="Pursonal_Budget"
android:roundIcon="@mipmap/ic_launcher_round"
android:supportsRtl="true"
android:theme="@style/Theme.Pursonal_Budget">
<activity android:name=".BudgetActivity"></activity>
<activity android:name=".HomeActivity" />
<activity android:name=".RegistrationActivity" />
<activity android:name=".LoginActivity" />
<activity android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN"
<category android:name="android.intent.category.LA
</intent-filter>
</activity>
</application>
</manifest>
manifest › application › activity
Merged Manifest
Shop
Today
Week
Month
Savings
My Cart
Today
Week
Month
Analytics
History

Experiment-12

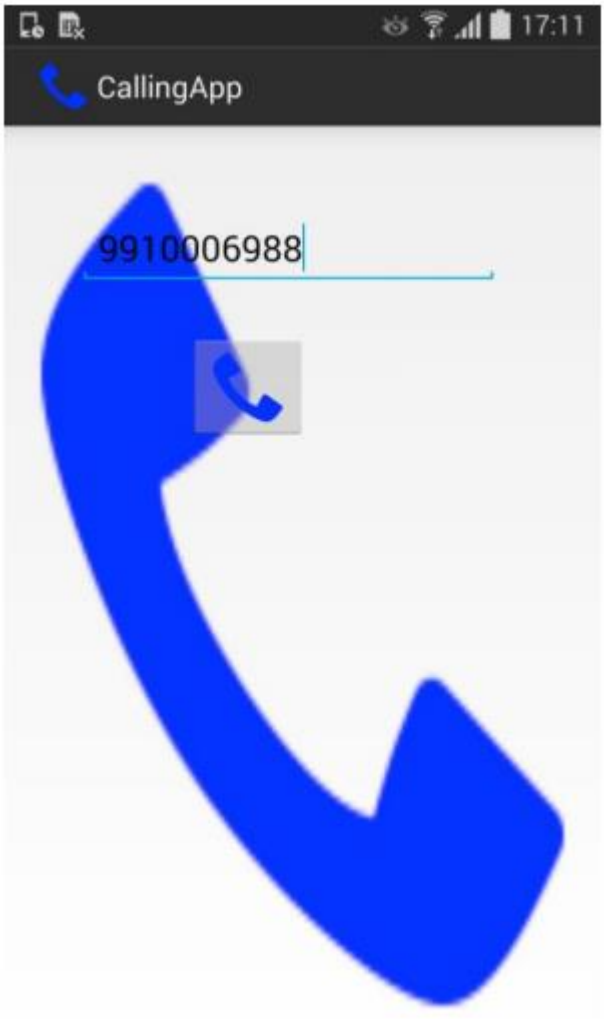
Aim: Program to develop a calling application.

Code:

```
package com.example.callingapp;
import android.net.Uri;
import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.view.Menu;
import android.view.View;
import
android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
import
android.widget.ImageButton;
public class CallActivity extends Activity implements
OnClickListener { EditText et1;
ImageButton bt1;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_call);
et1 = (EditText) findViewById(R.id.editText1);
bt1 = (ImageButton) findViewById(R.id.imageButton1);
bt1.setOnClickListener(this);
}
@Override
public boolean onCreateOptionsMenu(Menu menu) {
// Inflate the menu; this adds items to the action bar if it is present.
getMenuInflater().inflate(R.menu.call, menu);
return true;
}
@Override
public void onClick(View v) {
// TODO Auto-generated method stub
77
// Call a Number
String number = et1.getText().toString();
Intent call = new Intent(Intent.ACTION_CALL);
call.setData(Uri.parse("tel:" + number));
startActivity(call);
}
```

}

OUTPUT



Experiment-13

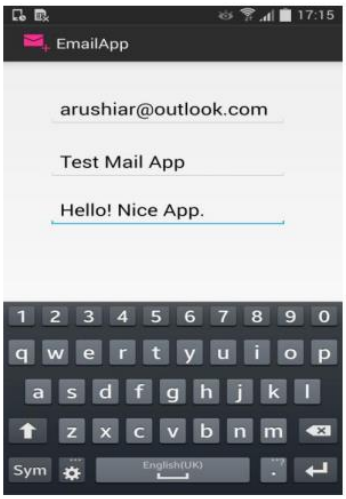
Aim: Program to develop a mailing application.

Code:

```
com.example.emailapp;
import android.os.Bundle;
import android.app.Activity;
import
android.content.Intent;
import android.view.Menu;
import android.view.View;
import
android.view.View.OnClickListener;
import android.widget.EditText;
import
android.widget.ImageButton;
import android.widget.Toast;
public class EmailActivity extends Activity implements
OnClickListener { EditText et1, et2, et3;
ImageButton ib1;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_email);
et1 = (EditText)
findViewById(R.id.editText1); et2 =
(EditText)
findViewById(R.id.editText2); et3 =
(EditText)
findViewById(R.id.editText3);
ib1 = (ImageButton) findViewById(R.id.imageButton1);
ib1.setOnClickListener(this);
}
@Override
public boolean onCreateOptionsMenu(Menu menu) {
79
// Inflate the menu; this adds items to the action bar if it is present.
getMenuInflater().inflate(R.menu.email, menu);
return true;
}
@Override
public void onClick(View v) {
// TODO Auto-generated method stub
```

```
String emailid = et1.getText().toString();
String[] emails = { emailid };
String subject = et2.getText().toString();
String message = et3.getText().toString();
Intent email = new Intent(Intent.ACTION_SEND);
email.setType("*/*");
email.putExtra(Intent.EXTRA_EMAIL, emails);
email.putExtra(Intent.EXTRA_SUBJECT, subject);
email.putExtra(Intent.EXTRA_TEXT, message);
startActivity(email);
Toast.makeText(this, "Mail Sent to Email Provider",
Toast.LENGTH_SHORT).show();
}
}
```

OUTPUT



Experiment-14

Aim: Program to design multiple activity class.

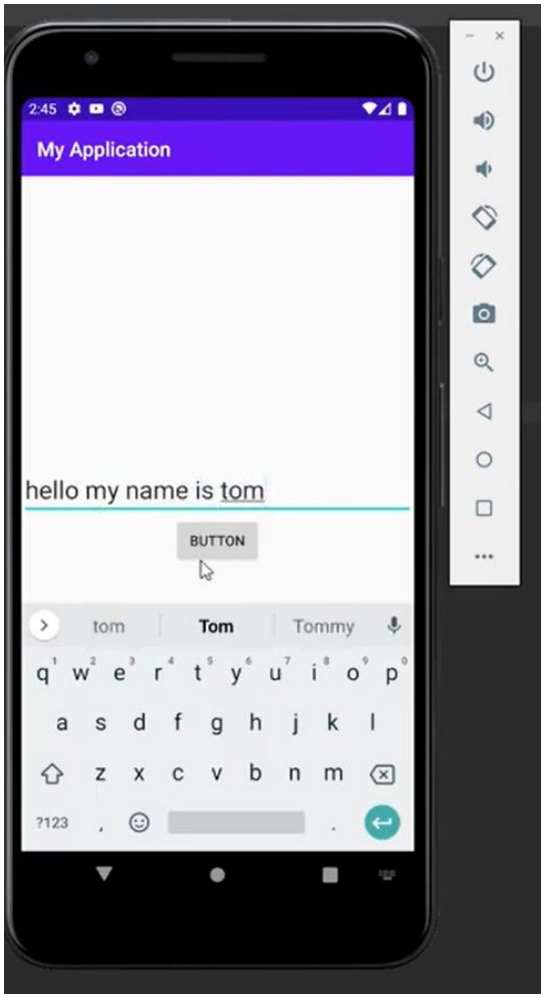
Code:

MainActivity.java

```
public class MainActivity2 extends AppCompatActivity {
    TextView textView,textView4;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        textView=findViewById(R.id.textView);
        Intent i=getIntent();
        String s=i.getStringExtra(MainActivity.EXTRA_NAME);
        textView.setText("Your name is : "+s);
        textView4=findViewById(R.id.textView4);
        String str=i.getStringExtra(MainActivity.EXTRA_NAME2);
        textView4.setText("Your Email is:\n "+str);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/editTextTextPersonName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:autofillHints=""
        android:ems="10"
        android:hint="Name"
        android:inputType="textPersonName"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
```

```
        app:layout_constraintVertical_bias="0.154" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:background="#AE4141"
        android:onClick="show"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.498"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/editTextTextPersonName"
        app:layout_constraintVertical_bias="0.208" />

    <ImageButton
        android:id="@+id/imageButton"
        android:layout_width="133dp"
        android:layout_height="123dp"
        android:onClick="help"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.816"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/button"
        app:layout_constraintVertical_bias="0.832"
        app:srcCompat="@drawable/th" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

OUTPUT